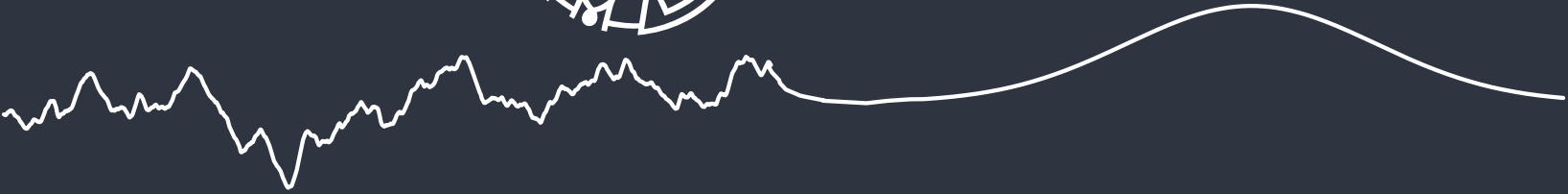




Universität
Marburg



Sitzung 8: Die Kluft zwischen Persönlichkeitsmaßen/Psychopathologie und den Neurowissenschaften

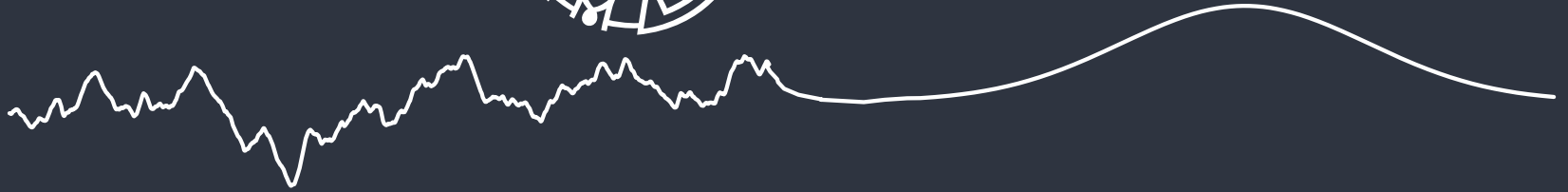
Dr. Philipp Bierwirth

Differentielle Psychologie und Persönlichkeitsforschung

SoSe 2026



Universität
Marburg



1. Introduction



Self-reported negative affect and neuroscience

Negative affect related traits (NARTs)

Negative Emotionality
Trait Anxiety
Negative dispositional affect
Harm avoidance
Neuroticism
Negative affectivity

- Valid predictors of overall life quality and the development of psychopathology
- Huge interest in the neurobiological foundations

Interindividual differences in experiencing negative affect are reflected by systematic interindividual differences in brain structure and function



Gruppenarbeit

Sie sind ein Team aus forschenden Diagnostikern und Psychologen, das von einem großen Forschungsinstitut rekrutiert wurde. Bisher verlässt sich das Institut bei der Diagnostik von Trait-Ängstlichkeit und Neurotizismus fast ausschließlich auf Selbstauskunft-Fragebögen (wie den NEO-PI-R oder das STAI). Das Problem: Fragebögen sind anfällig für soziale Erwünschtheit und mangelnde Introspektionsfähigkeit.

Ihre Mission: Das Institut möchte eine neue, objektive und multimodale "Diagnostik-Batterie" entwickeln. Ihre Aufgabe ist es, Ihr bisheriges Wissen über die Neurobiologie und Psychophysiologie zu nutzen, um fundierte biologische und physiologische Marker (Korrelate) vorzuschlagen, die Ängstlichkeit und Neurotizismus messbar machen.

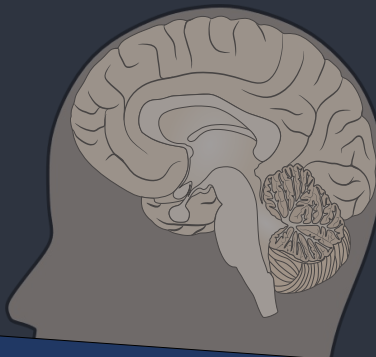
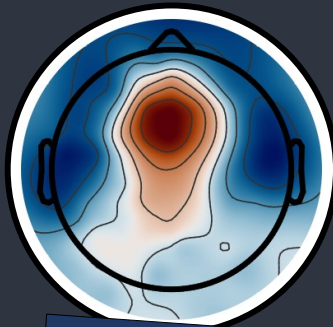
Arbeitsauftrag: Diskutieren Sie in Ihrer Forschungsgruppe (ca. 3-4 Personen) die folgenden Bereiche und entwerfen Sie eine diagnostische Testbatterie.



Neurobiological correlates of NARTs

Most promising candidates show **small effect sizes**, are **hard to replicate**, and often depend on **further moderating variables**:

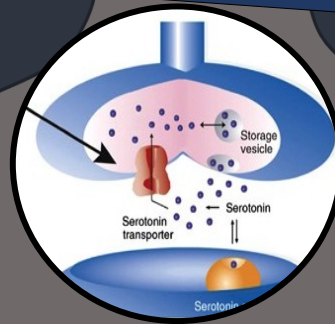
Increased **cognitive control-related** EEG activity (e.g., FM- θ , ERN, FRN & CRN)^{1,2}



Threat reactivity⁴
(e.g., heightened **amygdala** reactivity)

Reasons for the chasm between self-reported NARTs and neurobiological variables?

Interindividual differences in the **serotonergic system**³
(e.g., **5-HTTLPR**, serotonergic challenge tests)





Why is it so hard to link neurobiological variables to NARTs?

- Reasons that are responsible for the replication crisis (Researchers degrees of freedom, small sample sizes, etc.)
- Reasons that might be specific for personality neuroscience¹:



- 1: Neuropsychometrics
- 2: Size-assumption
- 3: Trait-scale semantics
- 4: Trait-hierarchy specification
- 5: Brain-trait theory
- 6: Abstraction-Level
- 7: Contextualized Trait
- 8: Context Taxonomy
- 9: Double-gap
- 10: Coherent reduction

1. Measurement of interindividual differences in neurobiological variables and trait variables
2. The hypothetical relationship between trait and neurobiological variables



Universität
Marburg

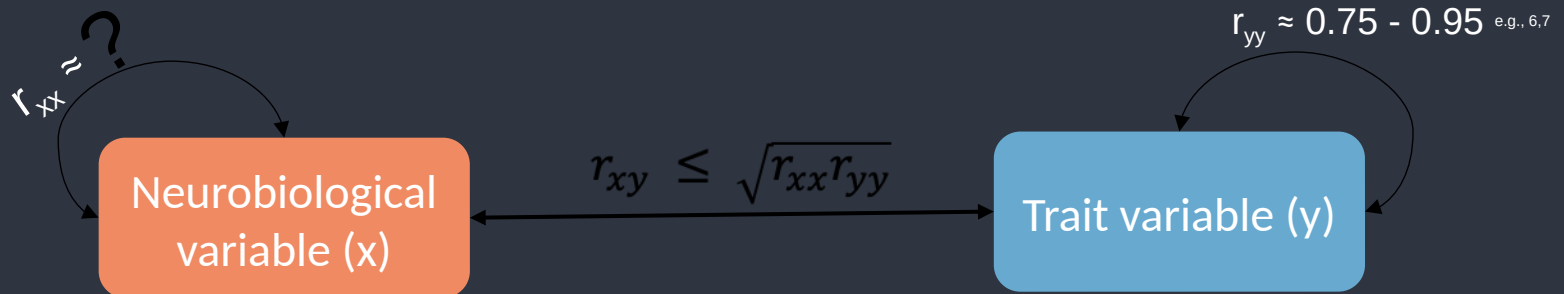


2. Measurement of interindividual differences in neurobiological and trait variables



Reliability of both measures

The reliability of each variable sets an upper limit for the maximum correlation between both variables:

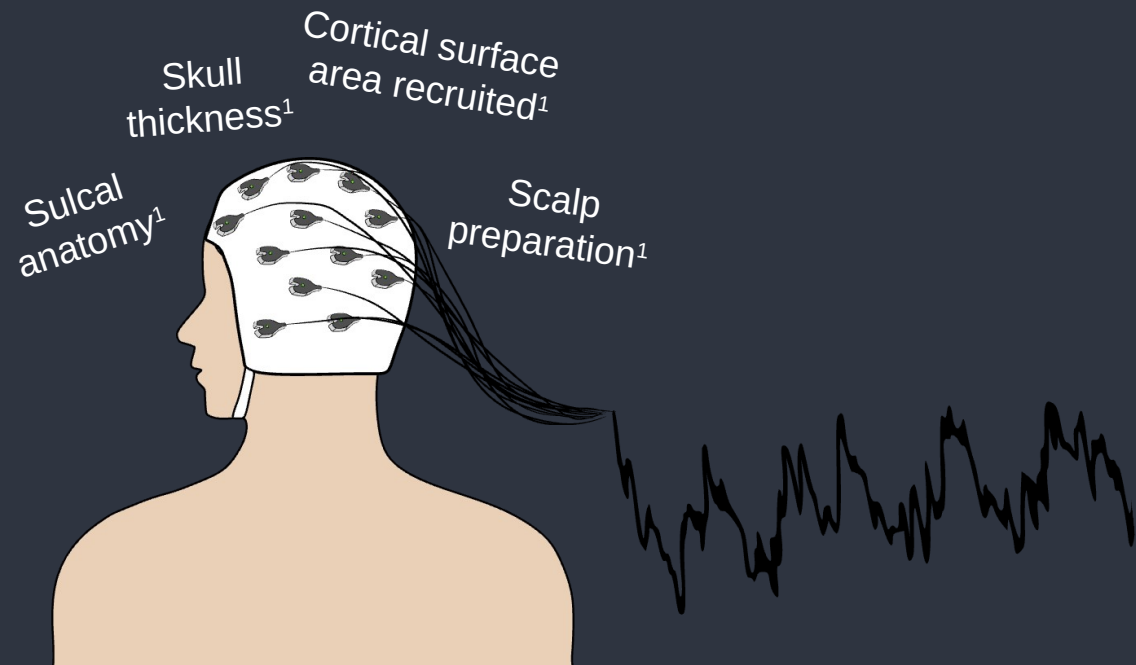


- Studies from our own lab showed moderate to high (retest-) reliabilities between $r_{xx} = \underline{0.78 - 0.98}$ for raw **N1 and P2 amplitudes**¹
- Other studies also reporting moderate to high reliabilities for raw ERP amplitudes: $\sim \underline{0.70 - 0.95}$ (e.g., ERN, FRN, LPP)^{e.g., 3,4,5}



Reliability of neurobiological variables

Factors that probably **increase** reliability,
but are **unrelated** to the neurocognitive process of interest:

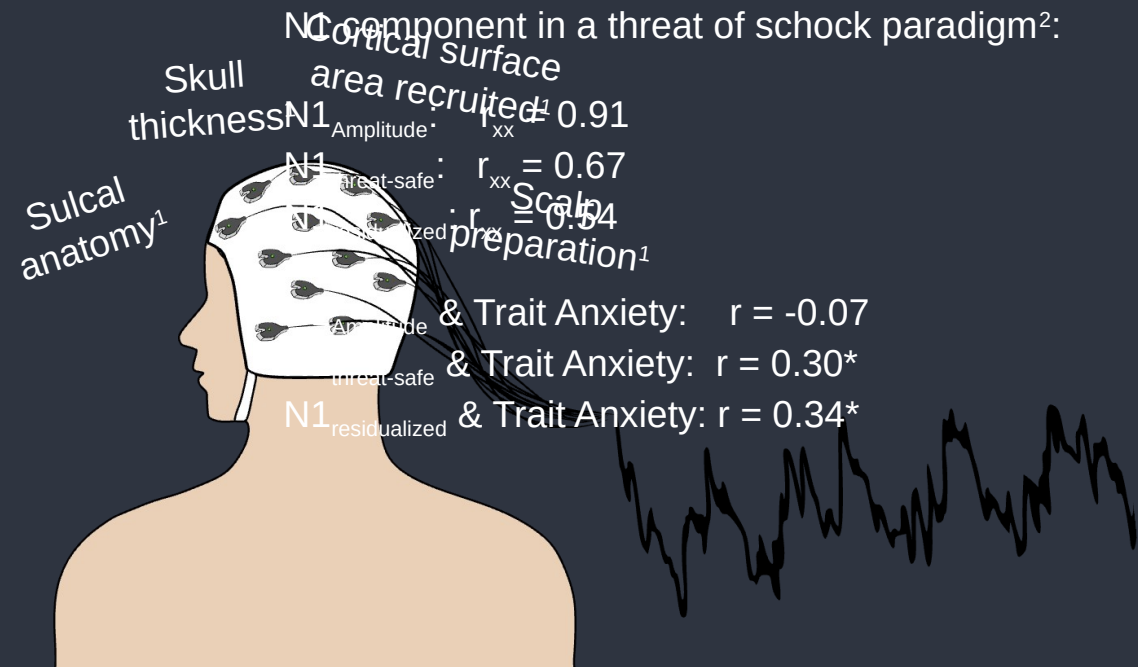


Systematic noise increases reliability but **decreases validity** of
(neurobiological) measurements.



Reliability of neurobiological variables

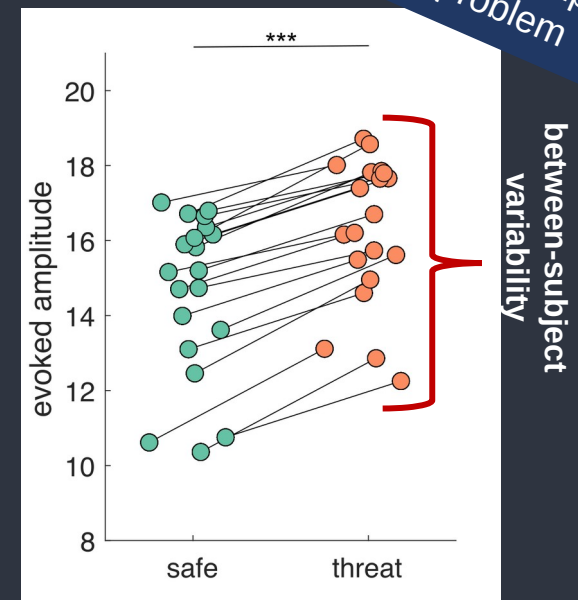
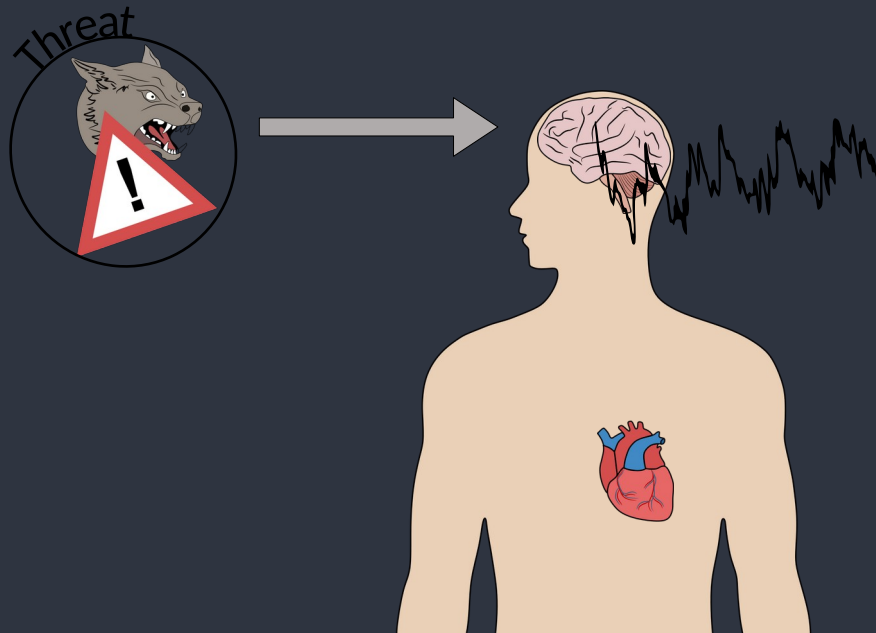
Factors that probably **increase reliability**,
but are **unrelated to the neurocognitive process** of interest:



Systematic noise increases reliability but **decreases validity** of
(neurobiological) measurements.

Validity of neurobiological measures

- Very **robust within-subject effects** of threat/feedback on FM- θ and heart rate
- But do these variables also capture valid **between-subject variability**?



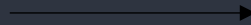


Validity of neurobiological measures

e.g., the size-assumption problem

Example:

- Baseline heart rate is correlated with the evoked heart response towards external stimuli (e.g., threats)



Resulting problem:

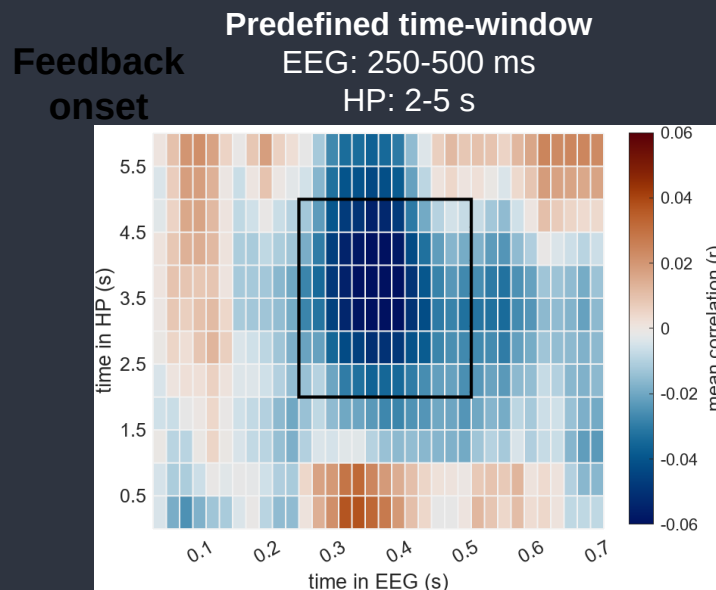
- Between-subject variability is influenced by physiological differences in the response system
- This between-subject noise makes it harder to detect meaningful correlations

Potential solutions:

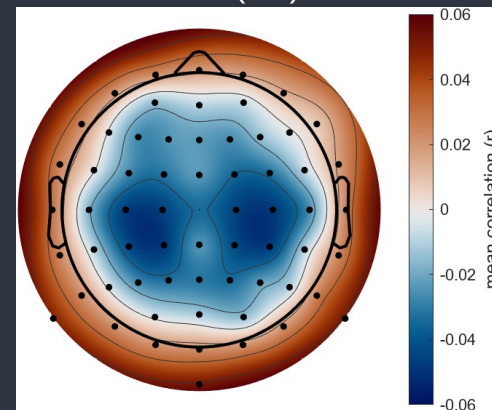
- Multivariate assessment of dependent neurobiological variables, e.g., via within-subject correlations or functional connectivity

Within-subject correlations to assess Brain-heart coupling

- One solution might involve the use of multivariate brain measures like **within-subject correlations** and **functional connectivity**
- **Example:** Cardio-electroencephalographic covariance tracing (CECT, Mueller et al. [2010]) is a multivariate measure that relies on within-subject correlations between single trial cardiac- and EEG-activity to assess functional brain-heart-coupling.



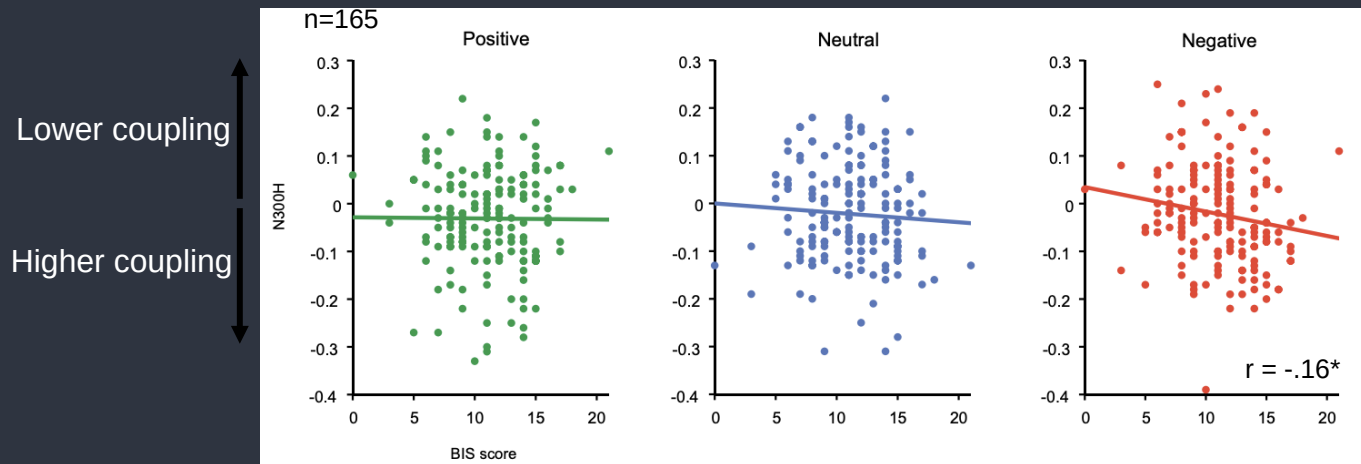
Scalp distribution of the N300H
average p-correlation within the
time window 250-500ms (EEG)
and 2-5s (HP)





Brain-heart coupling and trait anxiety

Studies using the CECT method showed that...



- ... **stronger brain-heart coupling** to negative feedback is associated with **trait anxiety** (Mueller et al., 2013; see Figure above)
- **panic disorder patients** show **elevated brain-heart coupling** (Mueller et al., 2014, Unterschemmann et al., 2024)

Thus, multivariate assessed brain-heart coupling might be a marker for trait and pathological anxiety....



Brain-heart coupling and trait anxiety

- ...in context of the multilab **CoScience** project¹, we aimed at replicating the observed association between trait anxiety and brain-heart coupling in **n = 662** subjects.
- Preliminary results showed **no association between the strength of brain-heart coupling and trait anxiety**



Bierwirth et al. (in prep)



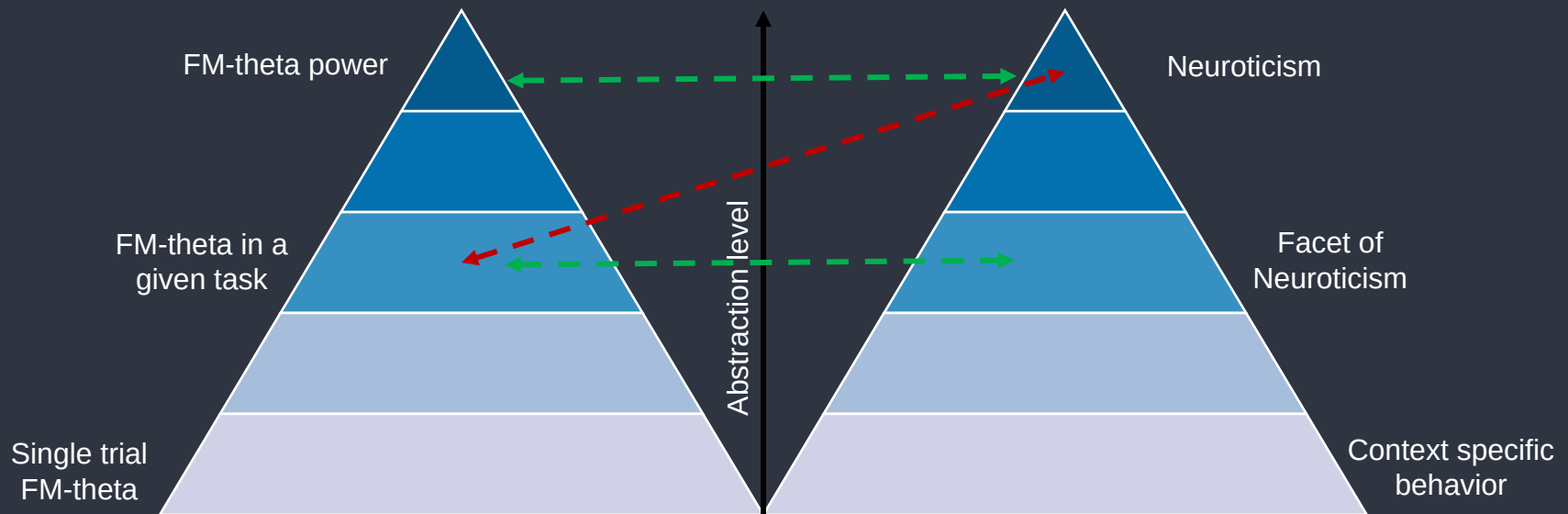
Universität
Marburg



3. The hypothetical relationship between trait and neurobiological variables



Abstraction level problem



Potential solutions:

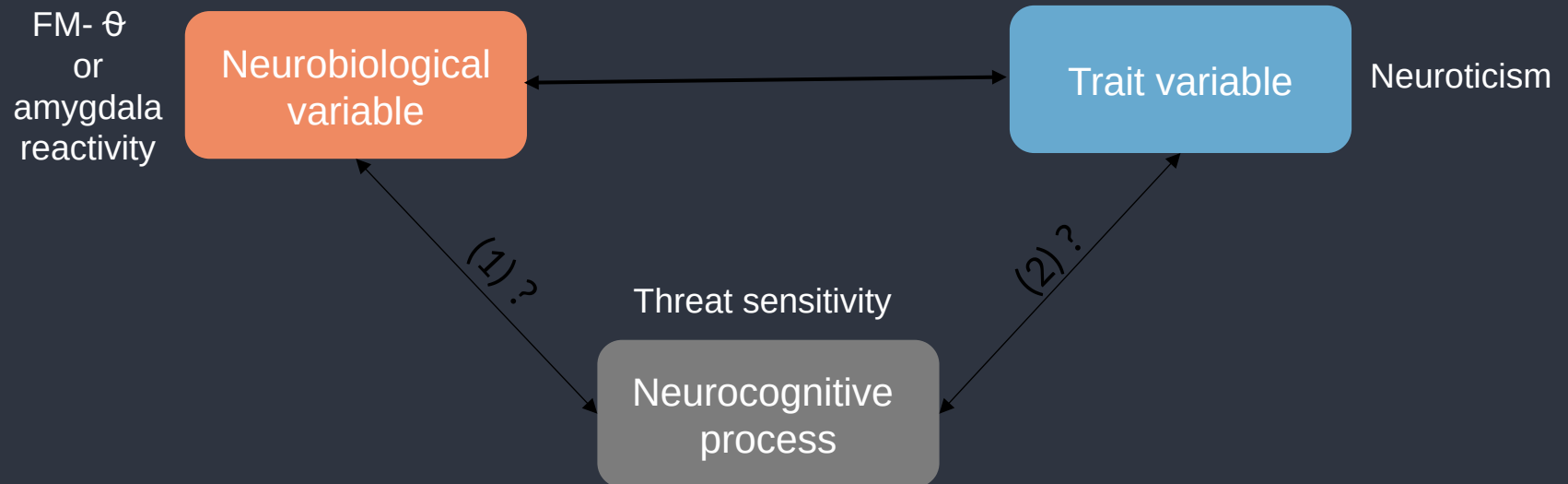
- Attempting to match abstraction level (e.g., threat-evoked theta power during fear conditioning and the neuroticism facet fearfulness)
- Aggregating neurobiological variables over different tasks to get a higher abstraction level (also reduces task specific noise)



The double-gap problem

Neurobiological personality research needs to fill two gaps:

1. Neurobiological variable – Neurocognitive process
2. Trait variable – Neurocognitive process

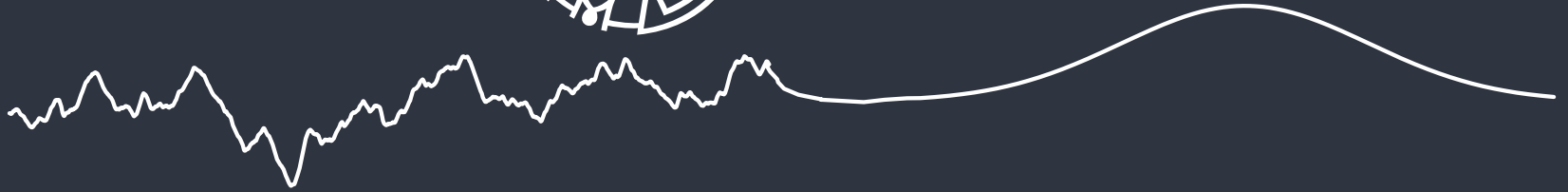


Potential solution:

Investigating one gap at a time to reduce uncertainties about the association between each variable and the neurocognitive process



Universität
Marburg



4. Summary



Summary

Acknowledging the presented (and other) problems, along with adopting good open science practices, could help detecting meaningful associations between neurobiological variables and NARTs.

